FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Advanced)-2020

PAPER – 2

Time Allotted: 3 Hours **Maximum Marks: 186**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

1. ***Section-A (01 – 08, 19 – 26, 37 – 44)*** *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Full Marks : **+4** If only (all) the four option(s) is (are) chosen.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen and both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–1** In all other cases.*

   Section-A (09 – 12, 27 – 30, 45 – 48) *contains 6 List-Match sets with 12 questions (each set has 2 questions). Each question has 4 statements in LIST-I & 5 or 6 statements in LIST-II. The codes for Lists have choices (A), (B), (C), (D) out of which only one is correct.*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-C (13 – 18, 31 – 36, 49 – 54)*** *contains 18 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

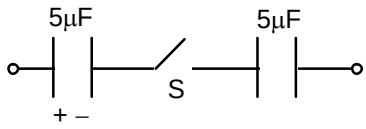
PART – I

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

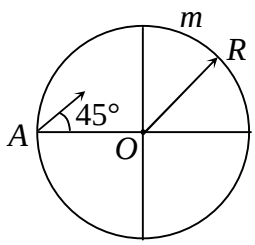
1. Let I_A and I_B be moments of inertia of a body about two axes A and B respectively. The axis A passes through the centre of mass of the body but B does not, then
(A) I_A may be greater than I_B (B) I_B may be greater than I_A
(C) I_A may be equal to I_B (D) I_A must be smaller than I_B

2. In the circuit shown, one of the capacitors is charged to 10V as shown by some external means and then connected to another uncharged capacitor as shown. After closing s, charge flown through 'S' (switch) will be



(A) 0 μC (B) 25 μC
(C) 50 μC (D) None of these

3. A ring of mass m and radius R is placed on a frictionless horizontal surface. A particle of mass m is projected from point A with velocity v at an angle of 45° with AO as shown. The correct statements are



(A) The particle reaches the same point A on the ring after time $\frac{4R\sqrt{2}}{v}$.

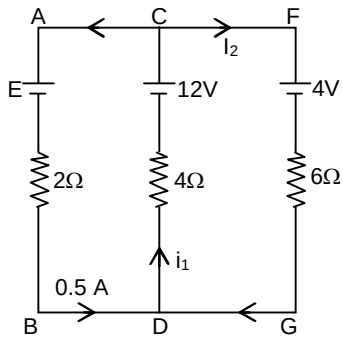
(B) Magnitude of impulse transformed during first collision is $\frac{mv}{\sqrt{2}}$.

(C) Magnitude of impulse transformed during second collision is $\frac{mv}{\sqrt{2}}$.

(D) Particle reaches diametrically opposite point on the ring in time $\frac{2R}{v}$.

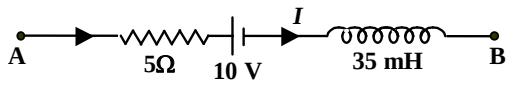
Space for Rough work

4. In the circuit shown in figure



(A) E = 6.6 V (B) i_1 = 1.1 A
(C) i_2 = 0.5 A (D) E = 4.4 V

5. The network shown in Figure is part of a complete circuit. If at a certain instant the current (I) is 5A, and is decreasing at a rate of 10^3 A/s, then find the value of $V_B - V_A$ in (V).

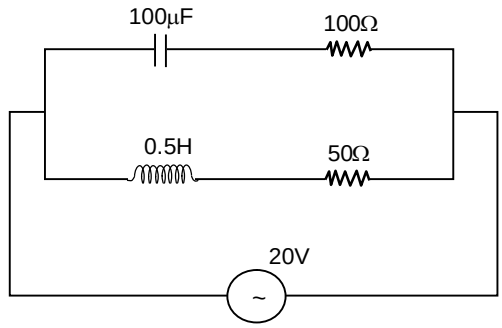


(A) 0 (B) 10
(C) 15 (D) 20

6. Using screw gauge the radius of the wire is found to be 2.50 mm. The length of wire is measured by using a scale and is found to be 50.0 cm. If mass of wire is measured as 25 g, then mark the correct statement(s) [Take π = 3.14]
(A) The density has to be computed upto 2 significant digit.
(B) The least count of scale used to measure length of wire is 1 mm.
(C) The density of wire is 2.5 g/cm^3.
(D) The least count of screw gauge is 0.01 mm

7. Light of wavelength 3500A^o is incident on two metals A and B, A of work function 4.2 eV and B of work function 1.19 eV respectively. The photoelectrons will be emitted by
(A) metal A (B) metal B
(C) both A and B (D) neither metal A nor metal B

8. In the given circuit, the AC source has ω = 100 rad/s. Considering the inductor and capacitor to be ideal, the



(A) the current through the circuit, I is 0.3 A
(B) the current through the circuit, I is $0.3\sqrt{2}$A
(C) the voltage across 100 Ω resistor = $10\sqrt{2}$ V
(D) the voltage across 50Ω resistor = 10 V

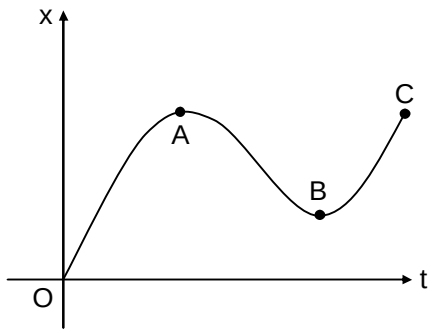
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(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO (02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

The displacement time (x – t) graph of a body acted by some forces is shown in the figure.



List – I		List – II	
(I)	From 'O' to just before 'A', the physical quantity of the body is	(P)	Zero
(II)	In between the points 'A' and 'B' the physical quantity of the body is	(Q)	Positive
(III)	In between the points 'B' and 'C' the physical quantity of the body is	(R)	Negative
(IV)	At the point 'A' the physical quantity of the body is	(S)	First negative then positive
		(T)	First positive then negative
		(U)	Cannot be explained.

9. If the physical quantity of the body is velocity then the correct match List-I and List-II.
(A) I $\rightarrow$ Q, II $\rightarrow$ R, III $\rightarrow$ Q, IV $\rightarrow$ P (B) I $\rightarrow$ R, II $\rightarrow$ Q, III $\rightarrow$ P, IV $\rightarrow$ Q
(C) I $\rightarrow$ U, II $\rightarrow$ R, III $\rightarrow$ Q, IV $\rightarrow$ P (D) I $\rightarrow$ T, II $\rightarrow$ U, III $\rightarrow$ S, IV $\rightarrow$ P

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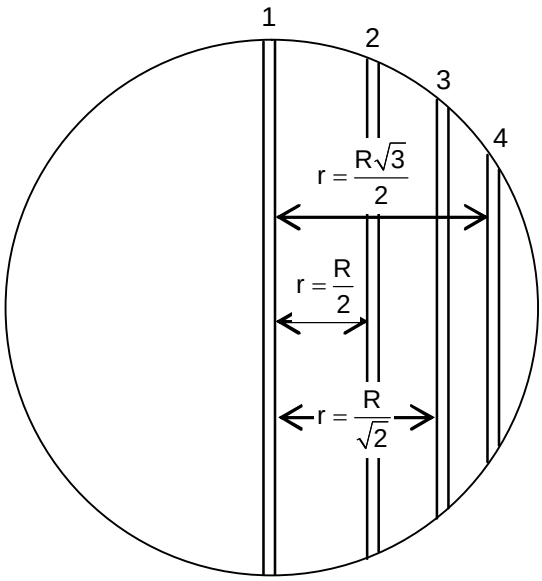
10. If the physical quantity of the body is acceleration then the correct match List-I and List-II.

(A) I $\rightarrow$ S, II $\rightarrow$ T, III $\rightarrow$ Q, IV $\rightarrow$ R (B) I $\rightarrow$ R, II $\rightarrow$ S, III $\rightarrow$ Q, IV $\rightarrow$ R

(C) I $\rightarrow$ T, II $\rightarrow$ S, III $\rightarrow$ Q, IV $\rightarrow$ P (D) I $\rightarrow$ S, II $\rightarrow$ Q, III $\rightarrow$ T, IV $\rightarrow$ P

Answer the following by appropriately matching the list based on the information given in the paragraph.

Consider a planet of radius R having density ρ. The four tunnels (1, 2, 3 and 4) are dug side at the distances from its centre are r = 0, $r = \frac{R}{2}$, $r = \frac{R}{\sqrt{2}}$ and $r = \frac{R\sqrt{3}}{2}$ respectively as shown. Four objects each of mass 1 kg are dropped inside the tunnel from the surface of plant at t = 0.



List-I gives the above four tunnels while List-II the magnitude of some quantity.

List – I		List – II	
(I)	Tunnel '1'	(P)	0
(II)	Tunnel '2'	(Q)	$\frac{2\pi G\rho R}{3}$
(III)	Tunnel '3'	(R)	$\frac{2\sqrt{2}\pi G\rho R}{3}$
(IV)	Tunnel '4'	(S)	$\frac{2\pi G\rho R}{\sqrt{3}}$
		(T)	$\frac{4\pi G\rho}{3}R$

Space for Rough work

11. The normal reaction between objects and mid of tunnels, the correct match is
(A) I → P, II → Q, III → R, IV → S (B) I → S, II → R, III → Q, IV → P
(C) I → Q, II → P, III → R, IV → S (D) I → P, II → Q, III → S, IV → R

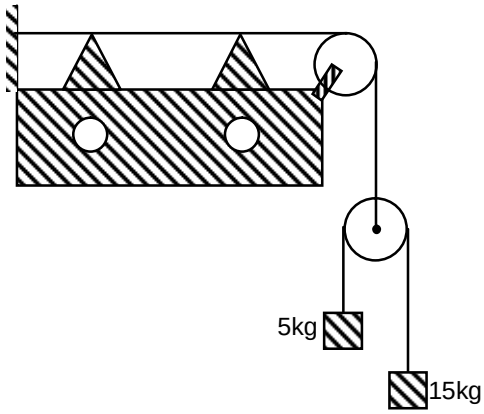
12. The magnitude of maximum acceleration of the objects in the tunnels
(A) I → T, II → S, III → R, IV → Q (B) I → S, II → T, III → R, IV → Q
(C) I → T, II → S, III → S, IV → R (D) I → Q, II → R, III → S, IV → T

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. XXXXX.XX).

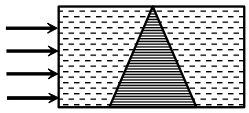
13. A proton is fired from very far away towards a nucleus with charge = 120 e, where e is the electronic charge. It makes a closest approach of 10 fm to the nucleus. The de Broglie wavelength (in units of fm) of the proton at its start is: (take the proton mass, $m_p = (5/3) \times 10^{-27}$ kg ; $h/e = 4.2 \times 10^{-15}$ J.s./C; $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ m/F; 1 fm = 10^{-15} m)

14. In a sonometer wire, When the tension is maintained by suspending a 20 kg mass from the free end of the wire, the fundamental frequency of vibration is 300 Hz. If the tension is provided by two masses of 5 kg and 15 kg suspended form a pulley (massless and friction less) as shown in the figure, if the fundamental frequency will approximately n65 Hz then find the value of 'n'



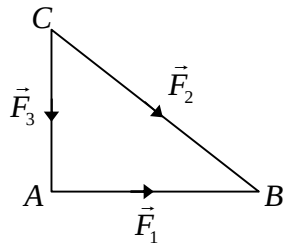
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15. A thin isosceles prism with angle 4^0 and refractive index 1.5 is placed inside a transparent tube with water $\left(\text{refractive index} = \frac{5}{4}\right)$ as shown. Find the deviation of light (in degree) due to prism



16. A particle is projected with speed u at an angle θ above horizontal. There are two times for which a projectile is at the same height. If the sum of these two times is equal to n $\frac{2u \sin\theta}{g}$, then find the value of 'n'.

17. Three forces $\vec{F}_1, \vec{F}_2$ and $\vec{F}_3$ starts acting on a particle of mass m. These forces are represented in magnitude and direction by three sides of triangle *ABC* as shown in figure. Find the acceleration of particle is $\frac{n\vec{F}_2}{m}$ then find the value of 'n'



18. An artificial satellite of mass m is moving in a circular orbit at a height h equal to radius of the earth from the earth's surface. Suddenly due to internal explosion the satellite breaks into two parts of equal masses. One part of the satellite stops just after the explosion and then falls to the surface of the earth. If find the increase in mechanical energy of the system (satellite + earth) due to explosion. (Acceleration due to gravity on the earth's surface is g) is n. (mgR) then find the value of 'n'.

Space for Rough work

Chemistry

PART – II

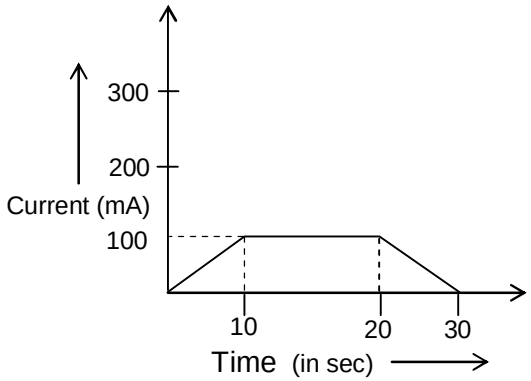
SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. A plot of ln $[A]_t$ vs t (where, $[A]_t$ is the concentration of the reactant at time t) for a reaction is linear with a negative slope of 0.0231 sec^{-1}. Then
(A) The initial concentration of the reactant will reduce to half in 30 seconds.
(B) A plot of $t_{1/2}$ vs concentration of reactant will give a straight line passing through origin.
(C) The reaction will be 90% complete in 100 seconds.
(D) The line in $ln[A]_t$ vs t plot will be steeper at higher temperature.

20. In which of the following the value of a (van der Waal's) constant of the left side gas is greater than that of the right side gas?
(A) SO_2, CH_4 (B) H_2, NH_3
(C) CO, N_2 (D) He, SO_2

21. When $CuSO_4$ solution is electrolysed in a copper voltameter for 30 sec, then 'm' gram of copper was deposited. The time-current graph for the electrolysis is:

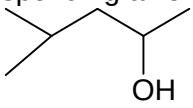
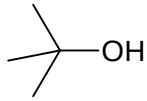
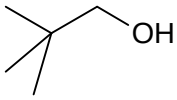
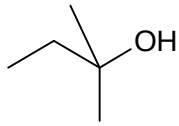


How much charge was passed through copper voltameter?
(A) 1 coulomb (B) 4 coulomb
(C) 2 coulomb (D) $\frac{1}{2}$ coulomb

Space for Rough work

22. Which of the following molecule(s) has/have zero dipole moment?
(A) CH_2Cl_2 (B) PCl_2Br_3
(C) XeF_6 (D) SF_6

23. $Al(OH)_3(s) \rightleftharpoons Al^{3+}(aq) + 3OH^-(aq)$
Which of the following substance(s) can increase the solubility of $Al(OH)_3$ according to above reaction?
(A) NH_4Cl (B) KCN
(C) $FeSO_4$ (D) KOH

24. Which of the following alcohol(s) can be prepared by acidic hydrolysis of the corresponding alkenes?
(A) 
(B) 
(C) 
(D) 

25. $\begin{pmatrix} CH_3CH_2COONa \\ CH_3COONa \end{pmatrix} \xrightarrow{\text{Electrolysis}} \text{Products}$
Which of the following product(s) is/are formed in the above reaction?
(A) C_2H_6 (B) C_3H_8
(C) C_4H_{10} (D) C_3H_6

26. Which of the following statement(s) is/are correct for the following complex?
$[Cr(NH_3)_4(Cl)_2]Br$
(A) It's coordination number is six
(B) It exhibits ionization isomerism
(C) The effective atomic number of chromium is 33
(D) It shows geometrical isomerism

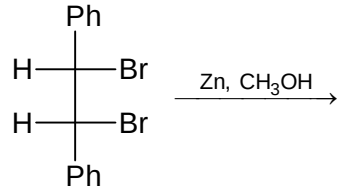
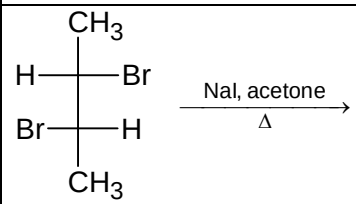
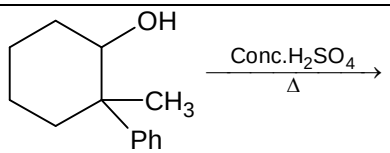
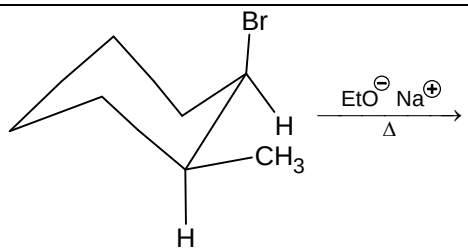
Space for Rough work

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO (02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

27. Answer the following by appropriately matching the lists.

List – I		List – II	
(I)		(P)	Anti elimination
(II)		(Q)	Rearrangement
(III)		(R)	Carbocation
(IV)		(S)	Transition state

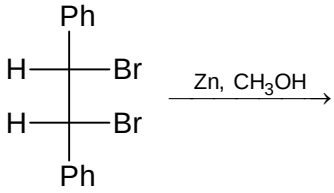
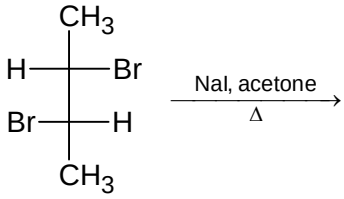
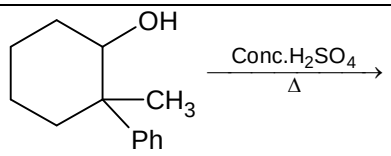
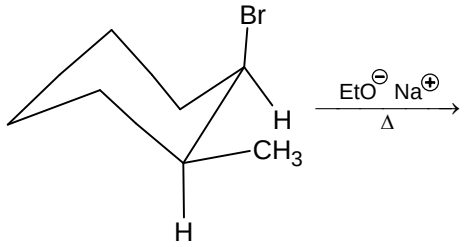
Which of the following options has the correct combination considering List-I and List-II?

(A) III→ QR (B) IV→ RS

(C) III→ PS (D) IV→ PR

Space for Rough work

28. Answer the following by appropriately matching the lists.

List – I		List – II	
(I)		(P)	Anti elimination
(II)		(Q)	Rearrangement
(III)		(R)	Carbocation
(IV)		(S)	Transition state

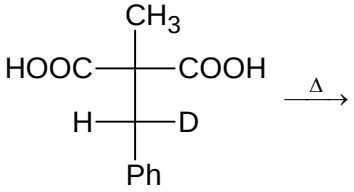
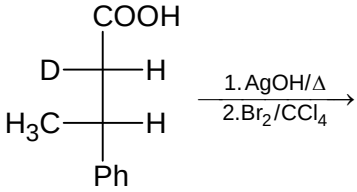
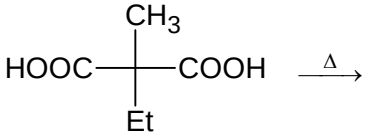
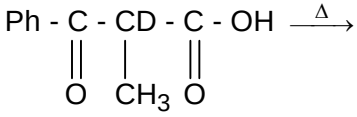
Which of the following options has the correct combination considering List-I and List-II?

(A) I $\rightarrow$ RS (B) II $\rightarrow$ QR

(C) I $\rightarrow$ PS (D) II $\rightarrow$ PQ

Space for Rough work

29. Answer the following by appropriately matching the lists.

List – I		List – II	
(I)		(P)	Diastereomers
(II)		(Q)	Racemic mixture
(III)		(R)	Optically active
(IV)		(S)	CO_2 will evolve

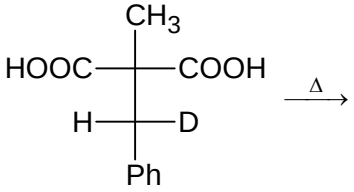
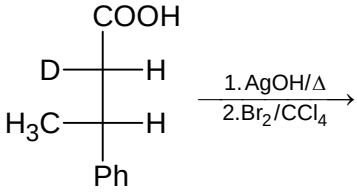
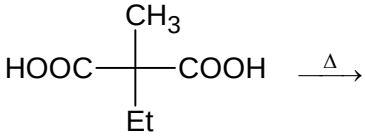
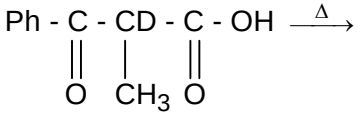
Which of the following options has the correct combination considering List-I and List-II?

(A) III→ QR (B) IV→ PS

(C) III→ QS (D) IV→ PR

Space for Rough work

30. Answer the following by appropriately matching the lists.

List – I		List – II	
(I)		(P)	Diastereomers
(II)		(Q)	Racemic mixture
(III)		(R)	Optically active
(IV)		(S)	CO_2 will evolve

Which of the following options has the correct combination considering List-I and List-II?

(A) I $\rightarrow$ QS (B) II $\rightarrow$ PRS

(C) I $\rightarrow$ PQR (D) II $\rightarrow$ PQ

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. XXXXX.XX).

31. How many elements of group-15 do not form pentachloride?

Space for Rough work

32. How many moles of copper atom(s) is/are deposited on the cathode if 8 faradays of electricity is passed through the aqueous solution of $CuSO_4$ which contains sufficient Cu^{2+} ions?

33. The average life of a radioactive element is 7.2 min. Calculate the time interval (in min) between the stags of 33.33% and 66.66% decay.

34. The radioactive decay $^{211}_{83}Bi \longrightarrow ^{207}_{81}Tl$ takes place in 100 L closed vessel at 27°C starting with 2 moles of $^{211}_{83}Bi\left(t_{1/2} = 130\,\text{sec}\right)$, the pressure developed in the vessel after 520 sec will be ____ atm

35. The measured voltage of the reaction with the indicated concentrations is 1.50 V. Calculate E°. $\left[Cr(s) + 3Ag^{+}(aq, 0.10\,M) \longrightarrow 3Ag(s) + Cr^{3+}(aq, 0.30\,M)\right]$

36. A solution containing H^+ and D^+ ions is in equilibrium with a mixture of H_2 and D_2 gases at 25°C. If the partial pressure of both gases are 1.0 atm, find the ratio of $[D^+]/[H^+]$
$\left[\text{Given } E^{o}_{D^+/D_2} = -0.003\,V\right]$

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. Two sides of a rhombus OABC (lying entirely in first quadrant or fourth quadrant) of area equal to 2 sq. units, are $y = \frac{x}{\sqrt{3}}$, $y = \sqrt{3}\,x$. Then possible coordinates of B is / are ('O' being the origin)
(A) $\left(1+\sqrt{3},\ 1+\sqrt{3}\right)$ (B) $\left(-1-\sqrt{3},\ -1-\sqrt{3}\right)$
(C) $\left(\sqrt{3}-1,\ \sqrt{3}-1\right)$ (D) none of these

38. Let 'z' be a complex number and 'a' be a real parameter such that $z^2 + az + a^2 = 0$, then
(A) locus of z is a pair of straight lines (B) locus of z is a circle
(C) $\arg(z) = \pm\frac{2\pi}{3}$ (D) $|z| = |a|$

39. $f(x) = \tan^{-1}(\sin x + \cos x)$, then f(x) is increasing in;
(A) $\left(-\frac{\pi}{2}, \frac{\pi}{4}\right)$ (B) $\left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$
(C) $\left(\frac{5\pi}{4}, \frac{3\pi}{2}\right)$ (D) $\left(-2\pi, -\frac{7\pi}{4}\right)$

40. Consider the planes $3x - 6y + 2z + 5 = 0$ and $4x - 12y + 3z = 3$. The plane $67x - 162y + 47z + 44 = 0$ bisects the angle between the given planes which
(A) contains the origin (B) is acute
(C) is obtuse (D) none of these

Space for Rough work

41. Probability if n heads in 2n tosses of a fair coin can be given by

(A) $\prod_{r=1}^{n}\left(\frac{2r-1}{2r}\right)$ (B) $\prod_{r=1}^{n}\left(\frac{n+r}{2r}\right)$

(C) $\sum_{r=0}^{n}\left(\frac{{}^{n}C_{r}}{2^{n}}\right)^{2}$ (D) $\frac{\sum_{r=0}^{n}\left({}^{n}C_{r}\right)^{2}}{\left(\sum_{r=0}^{2n}{}^{2n}C_{r}\right)}$

42. Let $f(n)=\begin{vmatrix} n & n+1 & n+2 \\ {}^{n}P_{n} & {}^{n+1}P_{n+1} & {}^{n+2}P_{n+2} \\ {}^{n}C_{n} & {}^{n+1}C_{n+1} & {}^{n+2}C_{n+2} \end{vmatrix}$ where the symbols have their usual meanings. Then f(n) is divisible by

(A) n^2+n+1 (B) $(n+1)!$

(C) $n!$ (D) none of these

43. $\int\sqrt{1+\operatorname{cosec} x}\,dx$ equals

(A) $2\sin^{-1}\sqrt{\sin x}+c$ (B) $\sqrt{2}\cos^{-1}\sqrt{\cos x}+c$

(C) $c-2\sin^{-1}(1-2\sin x)$ (D) $\cos^{-1}(1-2\sin x)+c$

44. Let $A=\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{bmatrix}$, then

(A) $A^2-4A-5I_3=O$ (B) $A^{-1}=\frac{1}{5}(A-4I_3)$

(C) A^3 is not invertible (D) A^2 is invertible

Space for Rough work

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO (02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

The chord AC of the parabola $y^2 = 4x$ subtends 90° angle at points B and D on the parabola. If A, B, C and D are represented by t_1, t_2, t_3 and t_4, then match the following

List–I		List–II	
(I)	$\left\|\frac{t_2 + t_4}{t_1 + t_3}\right\|$	(P)	–1
(II)	$\|t_2t_4 - t_1t_3\|$	(Q)	0
(III)	minimum value of $\|t_1 - t_3\|$	(R)	1
(IV)	The y-coordinate of the mid point of the points of intersection of the tangents at A, C and B, D is	(S)	4
		(T)	3
		(U)	4

45. Which of the following is the only correct combination?
(A) II→U (B) II→Q
(C) I→U (D) I→S

46. Which of the following is the only correct combination?
(A) IV→Q (B) III→R
(C) IV→U (D) I→P

Space for Rough work

Answer the following by appropriately matching the list based on the information given in the paragraph.

The number of real roots of

List–I		List–II	
(I)	$\sqrt{x+9}+\sqrt{x}=2.7$	(P)	0
(II)	$\frac{\sqrt{13-x^2}}{\lvert x+1\rvert}=1$	(Q)	1
(III)	$\sqrt{x+5}-\sqrt{x}=1$	(R)	2
(IV)	$\sqrt{x+2\sqrt{x-1}}-\sqrt{x-2\sqrt{x-1}}=2$	(S)	3
		(T)	4
		(U)	Infinite

47. Which of the following is the only correct combination?
(A) II→U (B) II→Q
(C) I→P (D) I→S

48. Which of the following is the only correct combination?
(A) IV→Q (B) III→R
(C) IV→U (D) I→P

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. XXXXX.XX).

49. If in a triangle ABC, b + c = 4a. Then $\tan\frac{B}{2}\tan\frac{C}{2}$ is equal to

Space for Rough work

50. Two rays of light coming along the lines y = 1 and y = –2 from the positive direction of x–axis and strikes a concave mirror whose intersection with the x – y plane is a parabola $y^2 = x$ at A and B respectively. The reflected rays pass through a fixed point C, then the area of the triangle ABC is

51. If $\cos 4x = a_0 + a_1 \cos^2 x + a_2 \cos^4 x$ is true for all values of $x \in R$, then the value of $\frac{18a_0 + a_1 + a_2}{2}$

52. $\lim\limits_{x\to 0} \frac{x\tan 2x - 2x\tan x}{(1-\cos 2x)^2} =$

53. If $\frac{\pi}{2} < \alpha < \frac{2\pi}{3}$ and $I = \int_0^{\sin 2\alpha} \frac{dx}{\sqrt{(4\cos^2\alpha - x^2)}}$ then the value of $\frac{(I+\alpha)}{5\pi}$ must be

54. If in a ΔABC, $BC = 5$, $CA = 4$, $AB = 3$ and D, E are the points on BC such BD = DE = EC, then $2\tan\theta$ is

Space for Rough work